

Reviewer Response Status — IEEE Access 33821

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Contents

All comments in one table
Reference numbers

12 comments, one-shot resubmission. One row per comment: what was asked, what changed, and what is still open. Manuscript sources: paper-access.tex (A) and paper-build.tex (B), kept in sync. Work line: main.

All comments in one table

Comment	Asked	Done	Remaining
R1-1 single collection, 31 images	Single dataset (CGHD-1152) and only 31 human-verified images; confidence intervals are wide; this cannot establish universal adaptation across drawing styles.	Results labeled as within-collection only. Drafter groups presented descriptively (16 direct and 15 inferred assignments disclosed, per-drafter table included): different hands give some style variation, but same corpus, so no universal-adaptation claim and no held-out-drafter experiment. Residual label bias disclosed as mitigated but not eliminated, with independently authored synthetic data as the second leg; future expansion described as future work.	Owner read-through. RE-VERIFY: 34-annotated to 31-scored accounting plus the 16-direct/15-inferred drafter chain (Bosco).
R1-2 complex parts	Complex devices (transformers, thyristors, optocouplers) are located but their pin relationships are not extracted, limiting supported components.	New capability table separating tested connectivity, estimated pins, and illustrative export, with a metric-boundary row; switch behavior stated exactly as the code does it (midpoint pins, conditional 0.001-ohm emission).	Owner read-through

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R1-3 hand-set thresholds	Scale-relative thresholds are manually tuned from median component diagonal; without adaptive learning, accuracy may decline for extreme sizes.	All values stated plainly (alpha 0.35 fallback-only with formula, multipliers 0.62/0.30/0.20, clamps, 4x reach, macro sweep 0.895-0.903); untested extremes admitted.	Nothing
R1-4 no values, can't simulate	No OCR/value-recognition loop; the output is topological rather than a complete simulatable SPICE netlist.	Scope stated in abstract, intro, and conclusion; retitled including heads and diagram; export labeled illustrative throughout, external values required.	Nothing
R1-5 small circuits only	The sample is allegedly dominated by 3-6-component circuits and lacks dense buses/multi-layer crossings; recall may drop on complex circuits.	Limits stated; true distribution printed in the caption (15 at most 5, 12 at least 10, median 7, range 3-14); nothing claimed beyond it.	Nothing
R1-6 memristor citations	Suggested citing two memristive SPICE-circuit papers (classical-conditioning neural circuit; decision-making circuit for navigation) in the SPICE-simulation sections.	Both checked and cited in Related Work with full references (IEEE Trans. Consumer Electronics, Feb 2026; IEEE Trans. Industrial Informatics, vol. 22).	Template check

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R2-1 statistics, roadmap	N=31 is statistically small. Clarify annotation cost and provide a dataset-expansion roadmap, or add rigorous statistical support; make conclusions conservative if this is the available data.	Sampling in four strata with independent annotation and adjudication described; roadmap added; missing timing honestly noted.	Owner read-through
R2-2 VLM comparison	VLM comparison has an oracle advantage because it receives ground-truth component boxes while the geometric pipeline supposedly bears detection errors. The comparison does not support a claim that geometry is superior in cost/ structural validity under those conditions.	Labeled as a diagnostic with identical annotated inputs (21/31 exact on both sides); no equivalence or cost claims.	Nothing
R2-3 title overclaims	Title says SPICE netlists while the paper says it does not read component values; reframe as topological netlist or correct the scope.	Retitled everywhere, including running heads and the pipeline diagram; metric boundary stated.	PDF rebuild
R2-4 crossing wires	Analyze the 70.7% crossover recall: missed crossover detection can merge electrically independent wires.	Measured tradeoff reported in full (27/4/0 goes to 19/0/8, stored artifact kept); removing edges also removes true connections, so no gain claimed; detector-miss part honestly marked unmeasured.	Nothing possible

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R2-5 ablations	Ablation study is severely insufficient.	Real table as observed, including fixed-pixel base 0.820 vs scale-relative 0.816 and full-pipeline ties at 0.890 presented as-is without spin; missing full-occlusion and scale tests disclosed.	Nothing provable
R2-6 single-scalar limit	Discuss the limitation of one scalar tau = kxs when component sizes vary within one schematic.	Kept as-is; no change needed.	Nothing

Shared language everywhere: a non-finding is not equivalence; no blanket guarantees; pair scores do not certify pin layouts; lab-condition results do not prove real-world dominance.

Reference numbers

Join 0.890 (418/37/66) on all 31; perfect-wire 0.88983 (420/40/64). Wire ledger: 0.9755 = 10°/18px config, 0.9726 = 12°/8px production. Otsu 0.789. VLM 0.923, 21/31 exact. Reach sweep macro 0.895–0.903. Numeric gate: 192/192 stored comparisons pass.